PIPELINE REPORT

Automated VQA Dataset Generation using Vision-Language Models

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Date: 7 July 2026

CONTENTS

Abstract

This work presents an automated pipeline for generating Visual Question Answering (VQA) datasets from print-media images using Vision-Language Models. The proposed system employs a two-stage generation strategy consisting of image description generation followed by category-specific question generation. Additional mechanisms such as category balancing, hallucination filtering, diversity enforcement, and output validation are incorporated to improve dataset quality. Experimental evaluation was conducted on a subset of a large-scale print-media image dataset, and Qwen2.5-VL-7B-Instruct was selected as the final model based on output quality and consistency.

1. INTRODUCTION

Objective:

The objective of this project is to develop an automated Visual Question Answering (VQA) dataset generation pipeline using Vision-Language Models (VLMs).The pipeline automatically generates multiple-choice question-answer pairs from images while ensuring diversity, grounding, and output quality.

Dataset Information:

Parameter	Value
Dataset Type	Print-Media Images
Total Available Images	~33,000
Images Downloaded for Experimentation	800
Images Used for Model Evaluation	50
Image Formats	JPG, PNG

The automated VQA generation pipeline was developed and evaluated on a print-media image dataset. The complete dataset contains approximately 33,000 images. Out of these, 800 images were downloaded for experimentation, and 50 representative samples were used for model comparison and pipeline validation.

Models Evaluated:

- Qwen2.5-VL-3B-Instruct
- Qwen2.5-VL-7B-Instruct
- Additional experimental model configurations

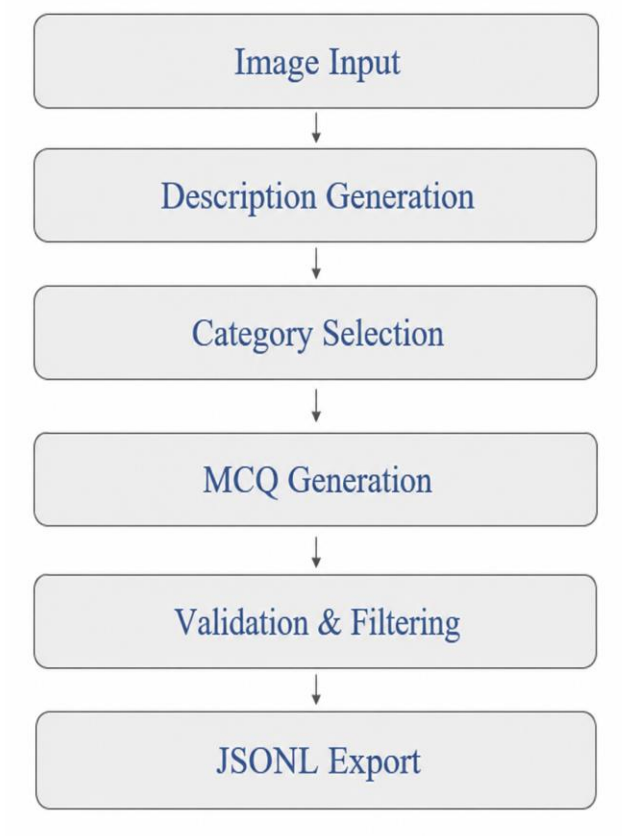
Final Selected Model:

- Qwen2.5-VL-7B-Instruct

Reason:

- Better grounding
- Better question diversity
- More consistent outputs

2. PIPELINE ARCHITECTURE



Pipeline Overview

The pipeline follows a two-stage generation strategy.

Stage 1

Generate a grounded image description.

Stage 2

Generate a category-specific multiple-choice question from the generated description.

This approach reduces hallucinations and improves question quality.

3. PIPELINE COMPONENTS

Step 1: Image Loading

Images are loaded from the print-media dataset directory and processed sequentially.

Example:

arc_0000008.jpg

The pipeline supports JPG, JPEG, and PNG image formats.

Step 2: Grounded Description Generation

A Vision-Language Model (Qwen2.5-VL-7B-Instruct) generates a detailed factual description of the image.

The generated description captures:

- Visible text
- Objects and entities
- Document layout
- Spatial arrangement
- Colors
- Countable elements

Example:

A computer screen displaying a table containing multiple rows and columns of data.

This description serves as the grounding context for question generation.

Step 3: Category Selection

To ensure diversity, the pipeline generates questions from multiple categories:

- OCR
- Recognition
- Localization
- Document Understanding
- Counting
- Colors
- Reasoning

A weighted deficit scheduling strategy is used to maintain a balanced distribution of question categories throughout dataset generation.

Step 4: Multiple-Choice Question Generation

Using the generated description and selected category, the model creates:

- One question
- Four answer options
- One correct answer

Example:

Question

Which of this best describes the content displayed on the computer screen?

Options

A. A list of items with detailed descriptions
B. A table with multiple rows and columns of data
C. A map showing geographical locations
D. A social media feed with user posts

Correct Answer

B

The generated output is required to follow a predefined JSON schema.

Step 5: Validation and Filtering

Before saving the output, multiple validation checks are performed:

Hallucination Filtering

Questions containing unsupported phrases such as:

"Cannot be determined"
“Not visible in the image”
“Unable to determine”

are rejected.

Duplicate Option Detection

Questions with repeated answer choices are discarded.

JSON Validation

Only properly formatted JSON outputs are accepted.

Diversity Enforcement

Recently used question starters are tracked to reduce repetitive question patterns.

Step 6: Dataset Export

Validated samples are stored in JSONL format.

Example:


```
{
 "id": "arc_0000008_q1",
 "question": "...",
 "correct answer": "B"
}
```


Implementation Note

The complete implementation of the pipeline is provided in the GitHub repository. Two primary modules were developed as part of this work:

1. image_download.py
Responsible for downloading and organizing print-media images from the provided source.

2. vqa_generation_pipeline.py
Responsible for image understanding, description generation, category selection, MCQ generation, validation, filtering, and JSONL dataset export.

These modules together form the complete end-to-end automated VQA dataset generation pipeline.

4. CHALLENGES AND SOLUTIONS

Problem 1: Repetitive Questions

Many generated questions started with similar patterns.

Example:

What is shown...
What is shown...
What is shown...

Solution

Starter diversity enforcement was introduced.

Problem 2: Hallucinated Information

The model occasionally generated information that was not present in the image.

Solution

Hallucination filtering patterns were added.

Problem 3: Category Imbalance

Some categories appeared more frequently than others.

Solution

Weighted category scheduling was implemented using target distributions.

Problem 4: Colab Session Interruptions

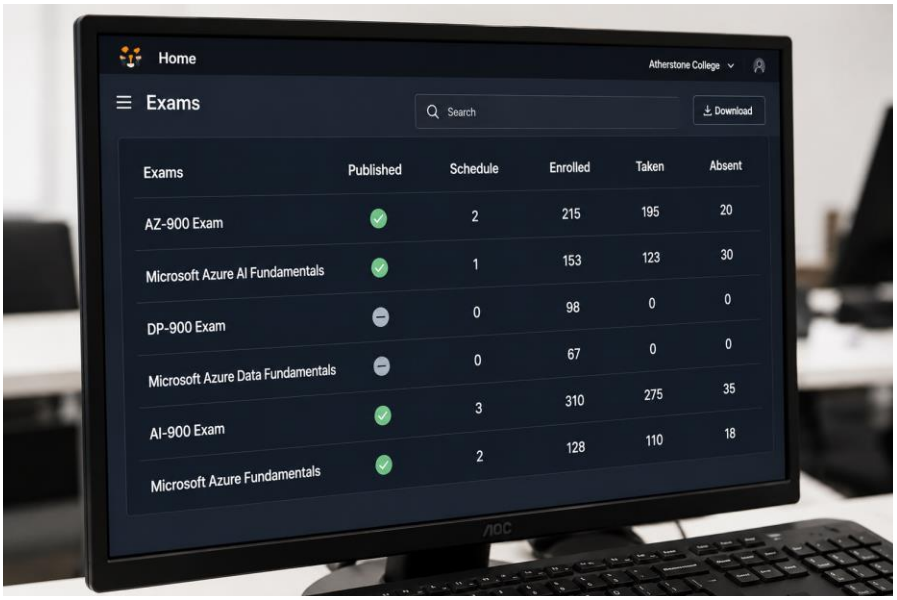
Long-running jobs could be interrupted.

Solution

progress. json

This allows the pipeline to resume from previous checkpoints.

5. INPUT EXAMPLE



Example:

arc_0000008.jpg

6. OUTPUT EXAMPLE

{
"id": "arc_0000008_q1",
"image_id": "arc_0000008",
"question": "Which of these best describes the content displayed on the computer screen?",
"option_a": "A list of items with detailed descriptions.",
"option_b": "A table with multiple rows and columns of data.",
"option_c": "A map showing geographical locations.",
"option_d": "A social media feed with user posts.",
"correct_answer": "B",
"qa_type": "recognition",
"category": "recognition",
"model_used": "Qwen/Qwen2.5-VL-7B-Instruct"
}

7.FINAL PIPELINE FEATURES

The final pipeline includes:

- Two-stage generation
- Category balancing
- Hallucination filtering
- Diversity-aware question generation
- Progress tracking
- JSONL export
- Hugging Face model compatibility
- Support for future API-based models

8. FUTURE IMPROVEMENTS

The current pipeline demonstrates automated VQA dataset generation using Vision-Language Models. Several enhancements can be incorporated in future versions:

• Integration of API-based multimodal models such as GPT-4o, Gemini, and Claude Vision.

• Support for dynamic model loading, enabling seamless switching between Hugging Face and API-based models.

• Generation of multiple question-answer pairs per image to increase dataset size and diversity.

• Automatic quality scoring and ranking of generated VQA samples.

• Addition of new question categories such as sentiment analysis, scene understanding, and chart/table reasoning.

• Distributed and large-scale processing for generating datasets from the complete image collection.

9. CONCLUSION

An automated VQA dataset generation pipeline was successfully developed using Vision-Language Models. The final system generates grounded and diverse multiple-choice question-answer pairs while reducing hallucinations and maintaining category balance. The pipeline can be extended further for large-scale dataset generation and support additional multimodal models.